

Statistical Analysis Report

Analysis Summary Template

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1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders.

Key Findings:

- Finding 1
- Finding 2
- Finding 3

2 Introduction

2.1 Background

Describe the context and motivation for the analysis. What problem are you trying to solve or what question are you trying to answer?

2.2 Objectives

Clearly state the objectives of this analysis:

1. Objective 1
2. Objective 2
3. Objective 3

2.3 Research Questions

- Research question 1?
- Research question 2?

3 Data

3.1 Data Source

Describe where the data comes from, how it was collected, and any relevant details about data provenance.

3.2 Data Description

```
# Load your data
# data <- read.csv("your_data.csv")

# For demonstration, using built-in dataset
data <- mtcars
```

3.2.1 Data Structure

```
# Display structure of the data
str(data)
```

```
'data.frame':  32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

3.2.2 Summary Statistics

```
# Summary statistics
summary(data)
```

mpg	cyl	disp	hp
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5
Median :19.20	Median :6.000	Median :196.3	Median :123.0
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0
drat	wt	qsec	vs
Min. :2.760	Min. :1.513	Min. :14.50	Min. :0.0000
1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000
Median :3.695	Median :3.325	Median :17.71	Median :0.0000
Mean :3.597	Mean :3.217	Mean :17.85	Mean :0.4375
3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000
Max. :4.930	Max. :5.424	Max. :22.90	Max. :1.0000
am	gear	carb	
Min. :0.0000	Min. :3.000	Min. :1.000	

1st Qu.:0.0000	1st Qu.:3.000	1st Qu.:2.000
Median :0.0000	Median :4.000	Median :2.000
Mean :0.4062	Mean :3.688	Mean :2.812
3rd Qu.:1.0000	3rd Qu.:4.000	3rd Qu.:4.000
Max. :1.0000	Max. :5.000	Max. :8.000

3.2.3 Sample of Data

```
# Display first few rows
head(data) %>%
  kable(caption = "Sample of the dataset")
```

Table 1: Sample of the dataset

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

3.3 Data Quality Assessment

3.3.1 Missing Values

```
# Check for missing values
missing_summary <- data %>%
  summarise(across(everything(), ~sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing_Count")

missing_summary %>%
  kable(caption = "Missing values by variable")
```

Table 2: Missing values by variable

Variable	Missing_Count
mpg	0
cyl	0
disp	0
hp	0
drat	0
wt	0
qsec	0
vs	0
am	0
gear	0
carb	0

3.3.2 Data Cleaning Steps

Describe any data cleaning, transformation, or preprocessing steps:

1. Step 1: Description
2. Step 2: Description

```
# Example data cleaning
# data_clean <- data %>%
#   filter(!is.na(variable)) %>%
#   mutate(new_variable = transformation)
```

4 Exploratory Data Analysis

4.1 Univariate Analysis

4.1.1 Numerical Variables

```
# Example: Histogram
data %>%
  ggplot(aes(x = mpg)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Miles per Gallon",
       x = "MPG",
```

```
y = "Frequency") +  
theme_minimal()
```

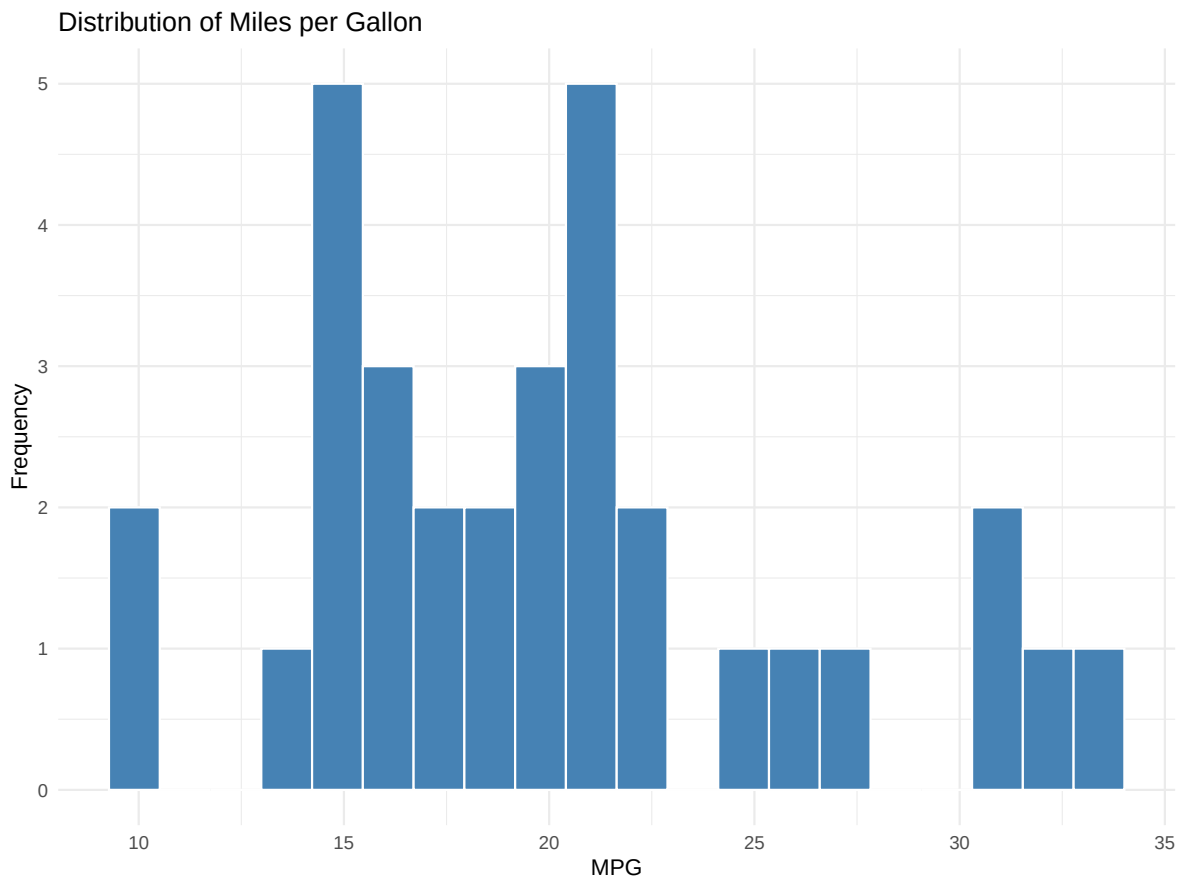


Figure 1: Distribution of numerical variables

4.1.2 Categorical Variables

```
# Example: Bar plot  
data %>%  
  ggplot(aes(x = factor(cyl))) +  
  geom_bar(fill = "coral") +  
  labs(title = "Distribution of Cylinders",  
        x = "Number of Cylinders",  
        y = "Count") +  
  theme_minimal()
```

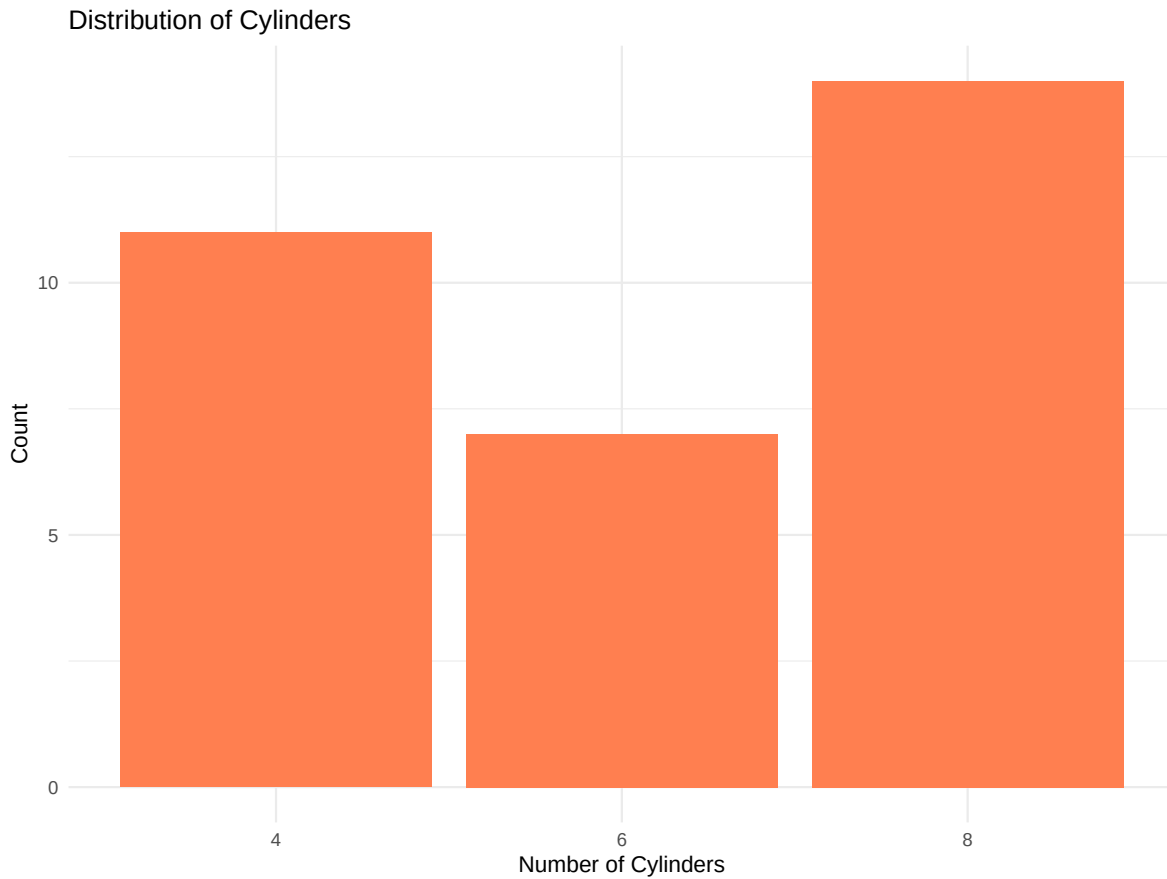


Figure 2: Distribution of categorical variables

4.2 Bivariate Analysis

4.2.1 Relationships Between Variables

```
# Example: Scatter plot
data %>%
  ggplot(aes(x = wt, y = mpg)) +
  geom_point(size = 3, alpha = 0.6, color = "darkblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(title = "Relationship between Weight and MPG",
       x = "Weight (1000 lbs)",
       y = "Miles per Gallon") +
  theme_minimal()
```

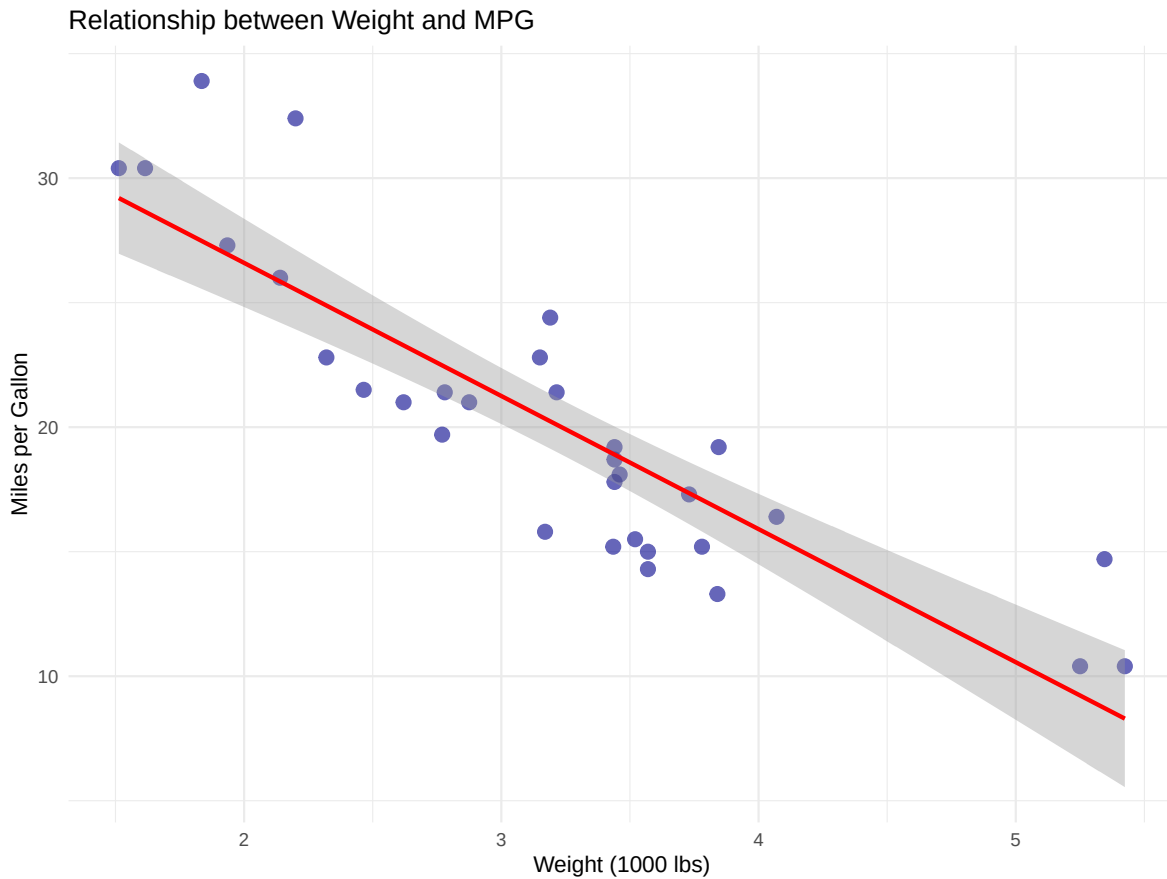


Figure 3: Relationship between variables

4.2.2 Correlation Analysis

```
# Correlation matrix
cor_matrix <- cor(data[, sapply(data, is.numeric)])

# Display correlation table
cor_matrix %>%
  kable(digits = 3, caption = "Correlation matrix")
```

Table 3: Correlation matrix

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
mpg	1.000	-	-	-	0.681	-	0.419	0.664	0.600	0.480	-
		0.852	0.848	0.776		0.868					0.551
cyl	-	1.000	0.902	0.832	-	0.782	-	-	-	-	0.527
		0.852			0.700		0.591	0.811	0.523	0.493	
disp	-	0.902	1.000	0.791	-	0.888	-	-	-	-	0.395
		0.848			0.710		0.434	0.710	0.591	0.556	
hp	-	0.832	0.791	1.000	-	0.659	-	-	-	-	0.750
		0.776			0.449		0.708	0.723	0.243	0.126	
drat	0.681	-	-	-	1.000	-	0.091	0.440	0.713	0.700	-
		0.700	0.710	0.449		0.712					0.091
wt	-	0.782	0.888	0.659	-	1.000	-	-	-	-	0.428
		0.868			0.712		0.175	0.555	0.692	0.583	
qsec	0.419	-	-	-	0.091	-	1.000	0.745	-	-	-
		0.591	0.434	0.708		0.175			0.230	0.213	0.656
vs	0.664	-	-	-	0.440	-	0.745	1.000	0.168	0.206	-
		0.811	0.710	0.723		0.555					0.570
am	0.600	-	-	-	0.713	-	-	0.168	1.000	0.794	0.058
		0.523	0.591	0.243		0.692	0.230				
gear	0.480	-	-	-	0.700	-	-	0.206	0.794	1.000	0.274
		0.493	0.556	0.126		0.583	0.213				
carb	-	0.527	0.395	0.750	-	0.428	-	-	0.058	0.274	1.000
	0.551				0.091		0.656	0.570			

Correlation matrix

5 Statistical Methods

5.1 Methodology

Describe the statistical methods and techniques used in this analysis:

- Method 1: Description and justification
- Method 2: Description and justification

5.2 Assumptions

List and verify any assumptions required for the statistical methods:

1. Assumption 1
2. Assumption 2

5.2.1 Checking Assumptions

```
# Example: Check normality
# qqnorm(data$variable)
# qqline(data$variable)
```

6 Results

6.1 Primary Analysis

6.1.1 Model Fitting

```
# Example: Linear regression
model <- lm(mpg ~ wt + hp + cyl, data = data)

# Model summary
summary(model)
```

Call:

```
lm(formula = mpg ~ wt + hp + cyl, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.9290	-1.5598	-0.5311	1.1850	5.8986

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	38.75179	1.78686	21.687	< 2e-16 ***
wt	-3.16697	0.74058	-4.276	0.000199 ***
hp	-0.01804	0.01188	-1.519	0.140015
cyl	-0.94162	0.55092	-1.709	0.098480 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.512 on 28 degrees of freedom
Multiple R-squared: 0.8431, Adjusted R-squared: 0.8263
F-statistic: 50.17 on 3 and 28 DF, p-value: 2.184e-11

6.1.2 Model Diagnostics

```
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model)
```

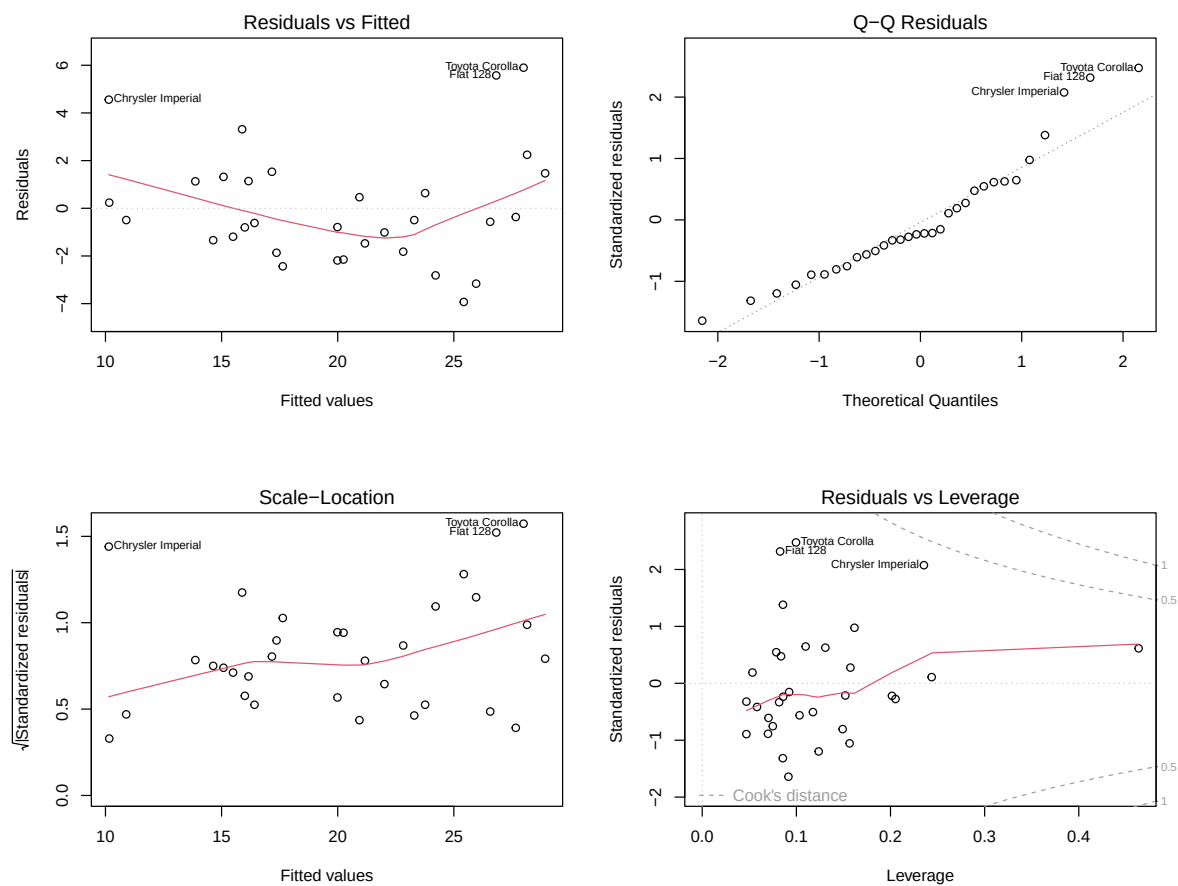


Figure 4: Model diagnostic plots

6.1.3 Results Summary

```
# Extract model coefficients
coef_table <- broom::tidy(model, conf.int = TRUE)

coef_table %>%
  kable(digits = 3, caption = "Model coefficients with 95% confidence intervals")
```

Table 4: Model coefficients with 95% confidence intervals

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	38.752	1.787	21.687	0.000	35.092	42.412
wt	-3.167	0.741	-4.276	0.000	-4.684	-1.650
hp	-0.018	0.012	-1.519	0.140	-0.042	0.006
cyl	-0.942	0.551	-1.709	0.098	-2.070	0.187

6.1.4 Interpretation

Interpret the results in context:

- **Variable 1:** Interpretation
- **Variable 2:** Interpretation

6.2 Secondary Analyses

6.2.1 Additional Tests

```
# Additional statistical tests
# Example: ANOVA
# anova_result <- aov(mpg ~ factor(cyl), data = data)
# summary(anova_result)
```

6.2.2 Subgroup Analysis

If applicable, present any subgroup analyses:

```
# Example: Analysis by subgroup
# data %>%
#   group_by(subgroup) %>%
#   summarise(mean_outcome = mean(outcome),
#             sd_outcome = sd(outcome))
```

6.3 Effect Sizes and Practical Significance

Discuss the practical significance of findings, not just statistical significance:

```
# Calculate effect sizes
# Example: R-squared
r_squared <- summary(model)$r.squared
adj_r_squared <- summary(model)$adj.r.squared

cat("R-squared:", round(r_squared, 3), "\n")
```

R-squared: 0.843

```
cat("Adjusted R-squared:", round(adj_r_squared, 3), "\n")
```

Adjusted R-squared: 0.826

7 Discussion

7.1 Key Findings

Summarize the main findings from the analysis:

1. Finding 1 with interpretation
2. Finding 2 with interpretation
3. Finding 3 with interpretation

7.2 Limitations

Discuss any limitations of the analysis:

- Limitation 1
- Limitation 2
- Limitation 3

7.3 Implications

What are the practical implications of these findings?

7.4 Future Directions

Suggest areas for future research or analysis:

- Direction 1
- Direction 2

8 Conclusions

Provide clear, concise conclusions based on the analysis. Address the original research questions and objectives.

9 References

List any references or citations used in the analysis.

10 Appendix

10.1 Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
```

```

[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
[1] knitr_1.50      lubridate_1.9.4 forcats_1.0.1  stringr_1.5.1
[5] dplyr_1.1.4     purrr_1.0.4     readr_2.1.5   tidyr_1.3.1
[9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0

loaded via a namespace (and not attached):
[1] Matrix_1.7-3      gtable_0.3.6      jsonlite_2.0.0    compiler_4.5.1
[5] tidyselect_1.2.1  splines_4.5.1     scales_1.4.0      yaml_2.3.10
[9] fastmap_1.2.0     lattice_0.22-7    R6_2.6.1          labeling_0.4.3
[13] generics_0.1.4    backports_1.5.0   pillar_1.10.2     RColorBrewer_1.1-3
[17] tzdb_0.5.0        rlang_1.1.6       broom_1.0.10      stringi_1.8.7
[21] xfun_0.52         timechange_0.3.0  cli_3.6.5         mgcv_1.9-3
[25] withr_3.0.2       magrittr_2.0.3    digest_0.6.37     grid_4.5.1
[29] hms_1.1.3         nlme_3.1-168      lifecycle_1.0.4   vctrs_0.6.5
[33] evaluate_1.0.4    glue_1.8.0        farver_2.1.2      rmarkdown_2.29
[37] tools_4.5.1       pkgconfig_2.0.3   htmltools_0.5.8.1

```

10.2 Additional Tables and Figures

Include any supplementary materials here.

```
# Additional code or analyses
```